NeonPine – manufacturing package

1. Fabrication outputs (`fab/`, generated from rev B with KiCad 10)

File	Content
`NeonPine-gerbers.zip`	Gerber X2 (F/B Cu, Mask, Silkscreen, Paste, Edge.Cuts) + Excellon drill + drill map + job file
`NeonPine-pos.csv`	Pick & place, both sides, mm (front: light sensor Q2 + touch pad; everything else SMD on the back)
`NeonPine-bom.csv`	Grouped BOM (value, footprint, qty). **Add MPN / LCSC columns before quoting**
`NeonPine.step`	3D assembly (lamp and inductor models from `NeonPine.3dshapes/`)
`NeonPine-cover-stand-gerbers.zip`	Rear cover + stand base (one file, two outlines): Edge.Cuts, mask, silkscreen (markings), NPTH drill. **No copper.**

Regenerate:

```
kicad-cli pcb export gerbers -o fab/gerber/ --layers
F.Cu,B.Cu,F.Mask,B.Mask,F.Silkscreen,B.Silkscreen,F.Paste,B.Paste,Edge.Cuts
NeonPine.kicad_pcb
kicad-cli pcb export drill -o fab/gerber/ --format excellon --generate-map --map-format
gerberx2 NeonPine.kicad_pcb
kicad-cli pcb export pos -o fab/NeonPine-pos.csv --format csv --units mm --side both --
exclude-dnp NeonPine.kicad_pcb
kicad-cli sch export bom -o fab/NeonPine-bom.csv --fields
'Reference,Value,Footprint,${QUANTITY},${DNP}' --group-by Value,Footprint NeonPine.kicad_sch
kicad-cli pcb export step -o fab/NeonPine.step --subst-models --force NeonPine.kicad_pcb
python3 tools/cover.py   # then: kicad-cli pcb export gerbers -o fab/cover/ --layers
F.Mask,B.Mask,F.Silkscreen,Edge.Cuts NeonPine-cover.kicad_pcb
```

2. PCB specification

Parameter	Value
Layers / material	2, FR-4 TG≥135, **UL 94 V-0** (ask the fab for the UL file number for the technical file)
Thickness / copper	1.6 mm / 1 oz (35 µm) both sides
Size	54.8 × 69.1 mm incl. 2 stand tabs, irregular outline (star / tiers), corners R 1.5 mm, routed; 3 × Ø2.2 NPTH
Min track / space	0.2 / 0.15 mm (default class); HV and HVBUS classes 0.3 mm / **0.5 mm clearance**
Min drill / via	0.3 mm / 0.6 mm pad (vias tented both sides)
Finish	**ENIG** (flat touch pads, nicer look)
Solder mask / silk	green or black mask, white silk, both sides

Parameter	Value
Panel	breakaway tabs only on the pot bottom edge **between** the stand tabs (keep the tree edges smooth)
Cover + stand PCB	FR-4 1.6 mm, **no copper**, black (or green) mask both sides, white silk on the outer face, Ø2.2 NPTH
Test	100 % electrical test (flying probe)

3. Assembly

1. **Back side SMT** (all SMD parts): solder paste, place, reflow with a SAC305 profile.
2. **Front THT:** light sensor **Q2 TEPT4400**, hand-soldered: the **collector goes in the square pad (pad 1)**, emitter in pad 2. Check the lead identification against the Vishay drawing (the leads are marked E/C), and seat the sensor flat (3 mm ball facing forward). J2 is **not a part**, just three pogo pads on the back for programming.
3. **Lamps (THT, hand or selective solder):** * Bend the leads 90° about 1.5 mm from the glass seal, using a bending jig. **Never bend at the glass.** * The **cathode** lead goes to pad 1, the left pad seen from the front, which is on the transistor (Qn collector) net. With DC drive only the cathode glows, so when forming the leads, turn the electrode pair so the cathode plate faces the viewer. * Insert, lay the body flat on the front, solder on the back, and trim the leads to ≤ 1.5 mm (they sit under the rear cover). * The star lamp LA1 is mounted standing up through the star. * **Fix each lamp with a dab of neutral-cure silicone** under the body (risk H8).
4. **Cleaning: mandatory.** Wash off flux residues (no-clean residue + 105 V + a 2 MΩ feedback divider → leakage and HV drift). Dry at 60 °C for 30 min.
5. **Program and run the end-of-line test (§4, §5) before the cover goes on** (the pogo pads J2 are under the cover).
6. **Rear cover:** press 3 × M2 × 5 mm hex standoffs (F-F) onto the main board with M2 × 4 screws from the front (holes H1–H3), put the cover on (markings outward) and fix it with 3 × M2 × 4 screws. Torque ≈ 0.2 N·m. **No cover = no shipment** (risk H1).
7. Apply the serial/batch label in the S/N field of the cover.
8. **Stand:** push the tree tabs (front row) and the cover tabs (rear row) into the base slots; a drop of cyanoacrylate or a snug press fit (slot width 1.7 mm, tab 1.6 mm).

4. Programming

- Programmer: WCH-LinkE (SWD single-wire), pads J2: `5V – SWIO – GND` (pin 1 = 5V, square pad).
- Command: `minichlink -w firmware.bin flash -b` (ch32v003fun) or WCH-LinkUtility.
- After flashing, read back the firmware version / CRC and record it against the serial number.

5. End-of-line test (per unit, ≈ 30 s)

Fixture: USB-C power through a current meter, pogo pins on J2, a light-tight hood with an "eye" photodiode per lamp (or a camera).

Step	Check	Limit
1	Power on at 5.0 V, MCU idle (boost off)	I_in < 10 mA
2	Firmware **self-test mode** (via SWD flag): ramp HV to 105 V, MCU reports the ADC reading	100–110 V (via ADC); I_in 60–150 mA with all lamps on
3	Each lamp in sequence (camera / photodiodes)	all 10 strike within 1 s, none dark, no wrong lamp (pin-map check)
4	Touch pad: the fixture touches TP2 (through the mask) with a grounded finger probe	firmware registers it
5	Light sensor: hood dark vs. LED on	ADC delta > threshold
6	HV off → discharge	MCU reports < 60 V within 2 s
7	After the cover is fitted: power on through USB, all lamps light; visual check of lamp bonding, cover screws, label	OK

Log: serial number, firmware CRC, measured currents and HV, and pass/fail → keep with the batch record (traceability, GPSR).

6. Traceability

- Serial / batch format: `NP-1 <rev> <YYWW>-<nnnn>` on the rear-cover label, also in the EOL test log.
- Keep the component lot codes per production batch (lamps especially, plus their radioactivity / RoHS declarations).

7. Manufacturing cost – small series (rev B, prices checked 2026-10-06)

Per finished unit, in **EUR excluding VAT**. USD prices are converted at 1 EUR = 1.15 USD. Assumptions: * **SMT:** JLCPCB "economic" PCBA on the back side. * **Hand work:** lamps, light sensor, cover and stand at ≈ €18/h (CZ). * **Delivery:** to CZ by courier.

7.1 Sources

Item	Source	Basis
Electronic BOM	LCSC / JLCPCB parts library (live query, 2026-10-06)	price break at qty × units; table §7.4
Neon lamps	**TME quote for NEON-4** (EUR incl. 21 % VAT: 10+ 0.287 · 30+ 0.207 · 100+ 0.173 · 600+ 0.129 · 2400+ 0.086)	**Proxy:** the design uses **NEON-3**, same family and size. **Get the NEON-3 quote** – NEON-4 is a 135 V DC type and is not usable on the 105 V rail
SMT assembly	JLCPCB economic PCBA fees: setup $8, stencil $1.5, **$3 per unique extended part**, $0.0016 per joint	148 SMT joints; **12 unique extended parts** → $45.5 per order + $0.24 per board

Item	Source	Basis
PCBs	estimate in the style of JLCPCB's calculator (2 layers, 1.6 mm, ENIG, 55 × 69 mm; cover/stand board without copper)	**estimate – run the online quote**
Mechanics, packaging, freight	estimates	

7.2 Unit cost (rev B BOM as designed, with the in-stock substitutes noted in §7.4)

Cost item	10 pcs	100 pcs	500 pcs	Notes
Electronic BOM excl. lamps	2.75	2.15	1.87	$3.17 / $2.47 / $2.15 – see §7.4
10 × neon lamps	1.43	1.07	0.71	TME 100+ / 600+ / 2400+ break, ex VAT (NEON-4 proxy)
Main PCB (ENIG)	2.61	0.35	0.24	≈ $30 / $40 / $140 per order
Cover + stand PCB	1.30	0.39	0.26	≈ $15 / $45 / $150 per order
SMT assembly (JLC)	4.17	0.60	0.29	$45.5 per order + $0.24 per board
Freight PCB + PCBA → CZ	1.74	0.30	0.12	≈ $20 / $35 / $70 per shipment
M2 standoffs ×3 + screws ×6	0.30	0.15	0.12	
Hand assembly	3.60	3.00	1.80	12 / 10 / 6 min (lamp bending jig at 500)
Programming + end-of-line test	0.45	0.36	0.30	1.5 / 1.2 / 1 min
Box, insert, printed manual	1.00	0.70	0.50	
Unit cost, as designed	**≈ 19.4**	**≈ 9.1**	**≈ 6.2**	EUR ex VAT
Unit cost with the cost-down in §7.3	**≈ 18.4**	**≈ 8.3**	**≈ 5.5**	

Import: 21 % CZ import VAT on the JLC shipment (recoverable for VAT payers); PCBs/PCBA are normally duty-free – check the HS code. At 10 pcs the fixed per-order costs dominate: assembly fees, PCB minimums and freight are ≈ €10 of the €19.

7.3 Findings from the price check (act before ordering)

#	Finding	Action
1	**L11 SWPA6028S331MT:** rated only **300 mA** (3.64 Ω) and **0 in stock**	Use **SWPA6045S331MT (C83453)**: 330 µH, 570 mA, 1.27 Ω, $0.05–0.09, in stock. It is 6 × 6 × 4.5 mm with the same land-pattern family and still fits under the 5 mm cover standoffs. Update the value in `tools/gen.py`
2	**C4 1 µF 250 V in 1210 does not exist at LCSC** (1210 only goes to 220 nF at 250 V)	Change the footprint to **1812** and use **C106152** (1 µF 250 V X7R, $0.19–0.38), or use 2 × 470 nF. Small re-place and re-route
3	**Q1 DMN15H310SE: 1 in stock at JLC**, $0.37–0.66	Drop-in SOT-223 alternative **CJT04N15 (C81507)**: 150 V, 160 mΩ at 10 V, V_th 2.5 V, $0.10–0.20, 11 k in stock. **Check**

#	Finding	Action
		R_DS(on) at V_GS = 5 V on the datasheet. Saves ≈ €0.25/unit
4	**J1 USB4125-GF-A** is the most expensive part ($0.58–0.84)	Cost-down option: **HC-TYPE-C-6P-01A (C2894893)**, 6-pin power-only, $0.03–0.05, in stock. Needs a footprint change. Saves ≈ €0.5–0.7/unit
5	Q2 TEPT4400 costs $0.31–0.42 (THT, hand-soldered)	Acceptable – it replaces the CdS LDR (RoHS). A cheaper SMD ambient-light sensor would have to go on the front, which means a second SMT side
6	12 extended parts × $3 = $36 per order	Irrelevant at 500 pcs; at 10 pcs it is 30 % of the assembly cost. Using JLC "basic" alternatives where possible (C3 22 µF) saves $3 each

7.4 Electronic BOM detail (LCSC price at the needed quantity, USD per piece)

Ref	Qty	LCSC	Part	Library	10 units	100 units	500 units
U1	1	C5187096	CH32V003F4P6	extended	0.288	0.224	0.163
Q11–Q20	10	C118745	MMBTA42	extended	0.029	0.019	0.017
Q1	1	C155224	DMN15H310SE-13 (alt. C81507 CJT04N15)	extended	0.535	0.410	0.374
Q2	1	C132178	TEPT4400	extended	0.421	0.341	0.306
L11	1	C83453	SWPA6045S331MT (see 7.3 #1)	extended	0.088	0.072	0.064
D1	1	C85620	ES1D-LTP	extended	0.029	0.029	0.023
D2	1	C283870	SMAJ100A	extended	0.035	0.035	0.028
D3	1	C284108	SMF5.0A	extended	0.019	0.019	0.017
F1	1	C75464	MF-NSMF050-2	extended	0.039	0.039	0.030
J1	1	C3151650	USB4125-GF-A (alt. C2894893)	extended	0.839	0.642	0.584
C4	1	C106152	1 µF 250 V X7R **1812** (see 7.3 #2)	extended	0.302	0.225	0.206
C3	1	C29277	22 µF 10 V X5R 0805	extended	0.082	0.059	0.048
C1	1	C19702	10 µF 10 V 0603	basic	0.032	0.032	0.025
C2, C6, C7	3	C14663	100 nF 0603	basic	0.012	0.012	0.007
C5	1	C57112	10 nF 0603	basic	0.011	0.011	0.009
FB1	1	C1002	600 Ω @ 100 MHz 0603	basic	0.013	0.013	0.010
R21–R30	10	C17470	150 k 0805	basic	0.005	0.004	0.003
R9, R41–R50	11	C31850	22 k 0603	basic	0.004	0.003	0.003
R5, R6	2	C22935	1 M 0603	basic	0.002	0.002	0.002
R3, R4	2	C23186	5.1 k 0603	basic	0.002	0.002	0.001

Ref	Qty	LCSC	Part	Library	10 units	100 units	500 units
R1, R2, R7, R11	4	basic	47 / 100 k / 39 k / 1 k 0603	basic	0.002	0.002	0.002
Sum per board					**$3.17**	**$2.47**	**$2.15**
With CJT04N15 + HC-TYPE-C-6P					*$2.04*	*$1.62*	*$1.34*

Lamps: 10 per board → 100 / 1 000 / 5 000 lamps per series. R12 and C8 (snubber) are DNP and not counted.

7.5 One-time costs (not in the unit cost)

Item	≈ EUR
EMC tests (EN 55015 / 61547 + FCC Part 15B), accredited lab	2 600–5 200
Safety evaluation (GPSR file, IEC 60598/62368 method)	900–2 600
WEEE / packaging EPR registrations, authorised representative	250–1 300 per country per year
EOL test fixture (pogo bed, photodiodes) + lamp-bending jig	250–700
Prototype round (rev B samples, 10 pcs)	≈ 200

At 500 units, the one-time costs add **≈ €9–17 per unit**, more than the unit cost itself. Compliance pays off from about 2 000 units (≈ €2.5–4.5 per unit).